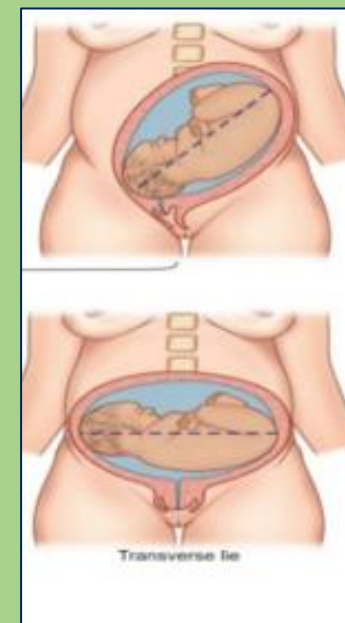


بسم الله الرحمن الرحيم

Breech presentation, transverse and oblique lie



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Learning objectives:

- **Definition of breech presentation.**
- **Causes of breech presentation.**
- **Risks associated with breech vaginal delivery.**
- **Management of women with breech presentation.**
- **Management of vaginal breech delivery.**
- **Definition of transverse and oblique lie.**
- **Causes of transverse and oblique lie.**
- **Management of transverse and oblique lie.**

Breech presentation:



-Is a type of mal-presentation when the presenting part is the buttock and lower limbs of the fetus.

Incidence :

- It is the commonest type of malpresentation (3-4%) at term.
- More in preterm delivery, with increasing gestational age, most of babies undergone spontaneous version to cephalic presentation.
- -Breech presentation occurs in 25% of births that occur before 28 weeks' gestation, declined to 3-4% at term.
- Perinatal mortality is increased 2-4 fold with breech presentation, regardless of the mode of delivery.
- Prematurity, cord accident and trauma are the main causes of morbidity and mortality with breech presentation.

Causes of breech presentation:

Same as other malpresentations:

1. Prematurity.
2. Uterine anomalies (septate uterus , unicornuate uterus)
3. Multiple pregnancy.
4. Amniotic fluid abnormalities (poly or oligohydramnios).
5. Fetal anomalies, especially hydro cephalus and anencephaly.
6. Placenta previa.
7. Extended fetal limbs prevent spontaneous version of the fetus to cephalic.
7. **Idiopathic**, most of cases no cause can be found.

Types of breech:



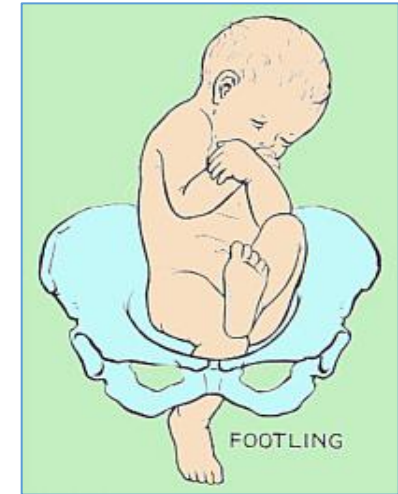
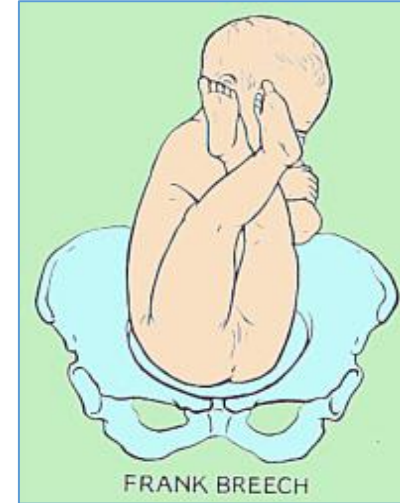
Frank breech 65%
Both legs extended



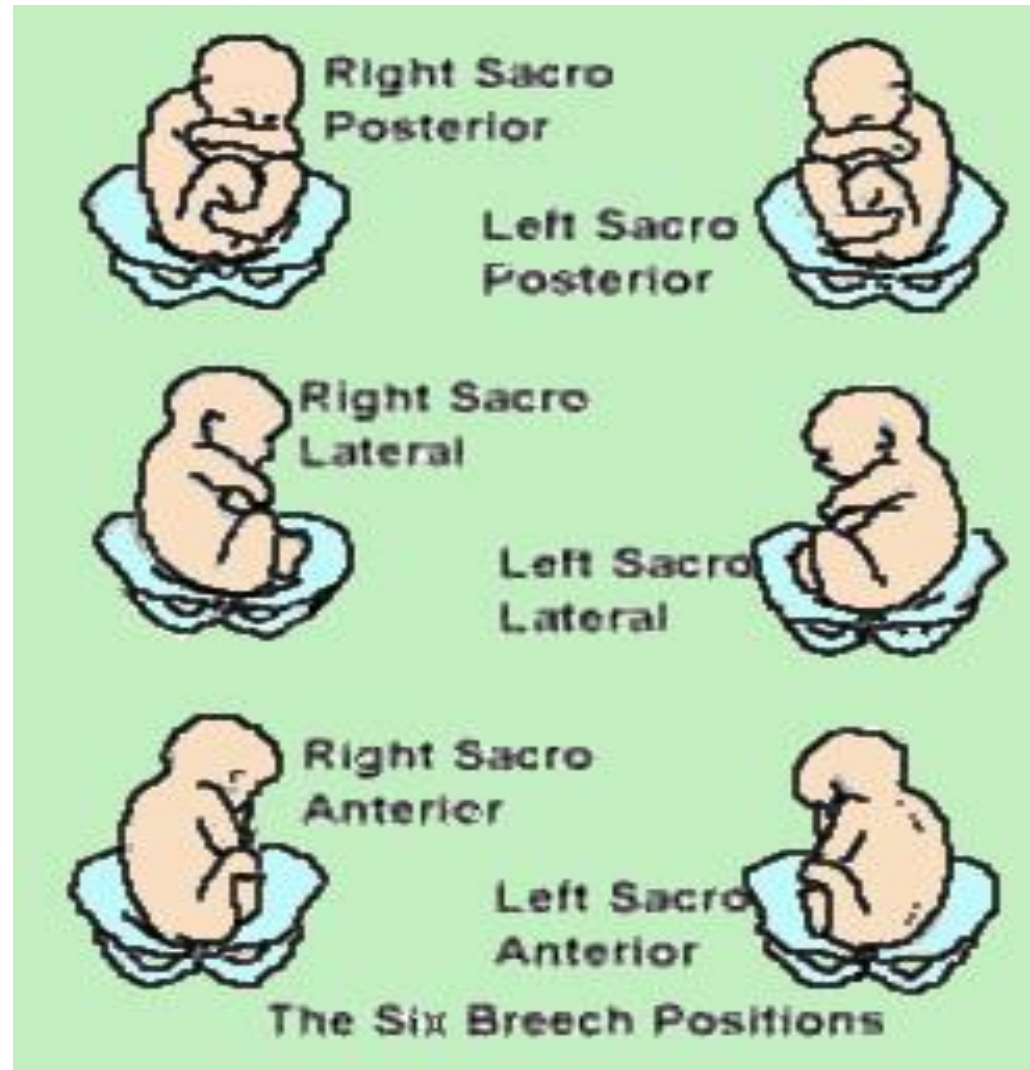
Complete breech 10%
Both legs flexed at knee
and hip



Footling breech 25%
One foot up, one foot down
(more common in multiparae
due to lax abdominal
muscles)



Position: depend on the position of sacrum of fetus.



Diagnosis

1. Abdominal examination.

- The fetal head palpable as a spherical hard mass at the fundus.
- The buttock which is broader & more soft is felt above or within the pelvis.
- The fetal heart sound: heard above the umbilicus.

2. Vaginal examination:

- We can feel the buttock which is soft.
- The anus (should be differentiated from the mouth in face presentation).
- The foot in case of complete breech.

3. Ultrasound: definitive diagnosis.

Antenatal management:



- The main concern with vaginal breech delivery is difficulty with delivering the after coming head.
- Although current literature recommends elective Caesarean section (C/S) for term breeches, training in assisted vaginal delivery is needed as some mothers elect to have assisted vaginal births.
- If breech presentation is diagnosed at any time in pregnancy before 36 weeks, nothing is done as most of them will undergo spontaneous version to cephalic presentation near term.

Management of persistent breech at 37 weeks:



- If a breech presentation persist after 36weeks:

-Three management options are available at this point and should be discussed with the woman.

- 1. External cephalic version.**
- 2. Allowing vaginal breech delivery.**
- 3. Elective Caesarean Section.**

To select the women for one of these options:

1. A proper obstetrical and medical history.

2. Abdominal palpation.

3. Ultrasound is performed at 37 weeks to confirm DX and assess:

- *The fetal birth weight (EBW)**

- *The biparietal diameter (BPD).**

- *The degree of extension or flexion of the head and legs (flexed head is good for vaginal delivery).**

- *The type of breech (Footling breech is not for vaginal delivery).**

- *Amount of amniotic fluid.**

- *Placental site.**

- *Any fetal anomaly not detected previously.**

4. Assessment of the size of the pelvis (clinically, sometime need X-ray or MRI).

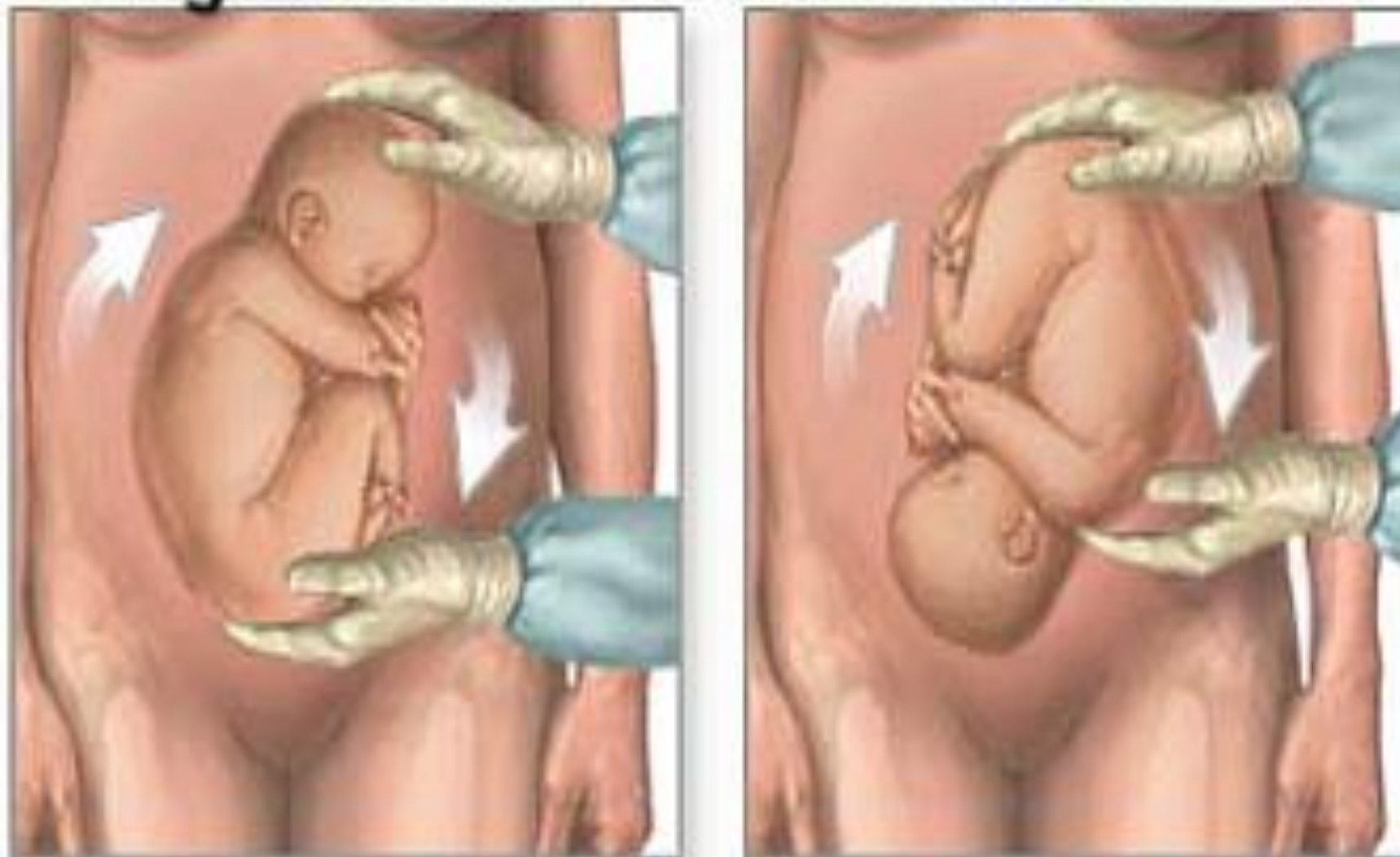
- **If assessment revealed an indication for C/S : the women should be scheduled for elective C/S e.g:**

- 1.Placenta previa.**
- 2. Medical disease with pregnancy.**
- 3. History of infertility.**
- 4. IUGR.**
- 5. Previous delivery by C/S.**
- 6.Old primigravida.**
- 7.Bad obstetric history ...ect.**

External cephalic version (ECV):

- This procedure is done by manipulating the fetus through the maternal abdomen and trying to turn it into cephalic presentation.
- ECV is relatively safe technique and has been shown to reduce the number of C/S due to breech.
- The success rates varies according to the experience of the operator(around 50% and are higher in Multiparous women).
- The woman should be carefully selected as there many contraindications and risks to this procedure.

Using external version to turn breech



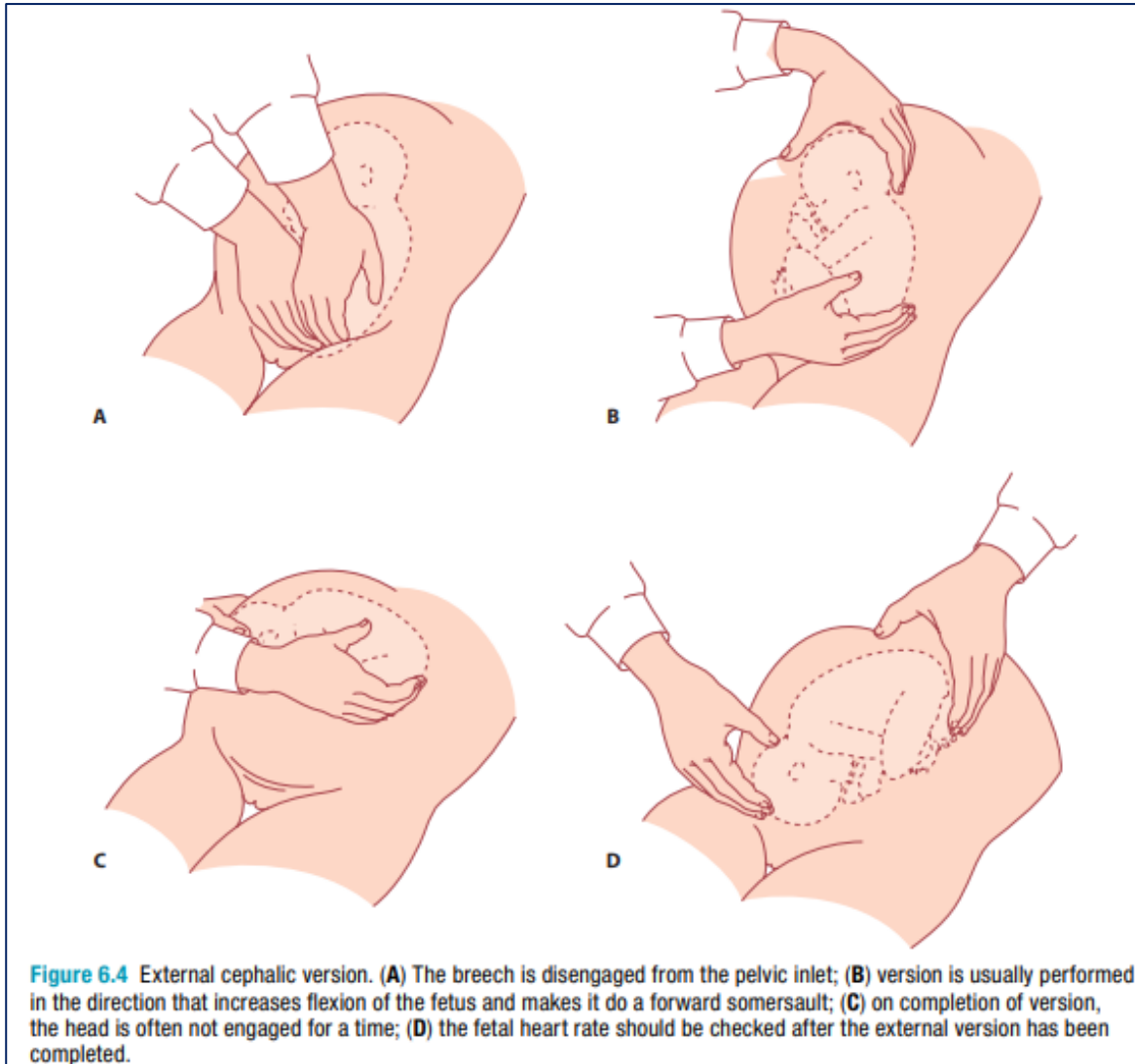
Factors which favour the success of ECV:

- Multiparty.
- Flexed breech presentation.
- Adequate liquor volume.
- Breech mobile above the pelvic brim.

- To increase the success rates of EVC:

- Trendelenburg position.
- Intravenous hydration.
- Uterine relaxation with tocolytic (e.g. Nifedipine)
- Give anti-D injection after ECV to Rh-ve woman.

External cephalic version



Contraindications to ECV:

- Fetal abnormality (e.g. hydrocephalus).
- Placenta praevia.
- Oligohydramnios or polyhydramnios.
- History of antepartum haemorrhage.
- Previous caesarean or myomectomy scar on the uterus.
- Multiple gestation.
- Pre-eclampsia or hypertension.
- IUGR
- Plan to deliver by caesarean section anyway

Risks of ECV:

1. Premature rupture of membranes.
2. Placental abruption.
3. Cord accident.
4. Transplacental haemorrhage in Rh-ve woman.
5. Rupture of uterus in the presence of previous scar in the uterus.
6. Fetal compromise or death.

Before & After doing ECV:

The CTG for 30–40 min should show a reactive normal trace and no uterine irritability.

-After the ECV, there should be no bleeding, leaking of amniotic fluid per vagina or uterine tenderness prior to discharge.

Persistent breech at term:

- If ECV fails or is contraindicated a decision has to be made on the mode of delivery - either by elective caesarean section or allowing vaginal delivery.
- The data for term breeches showed that perinatal morbidity is three times higher in the group delivered vaginally compared to those delivered by elective caesarean section.
- Therefore, each case must be judged carefully by an experienced obstetrician before a decision is made to allow a vaginal breech delivery.

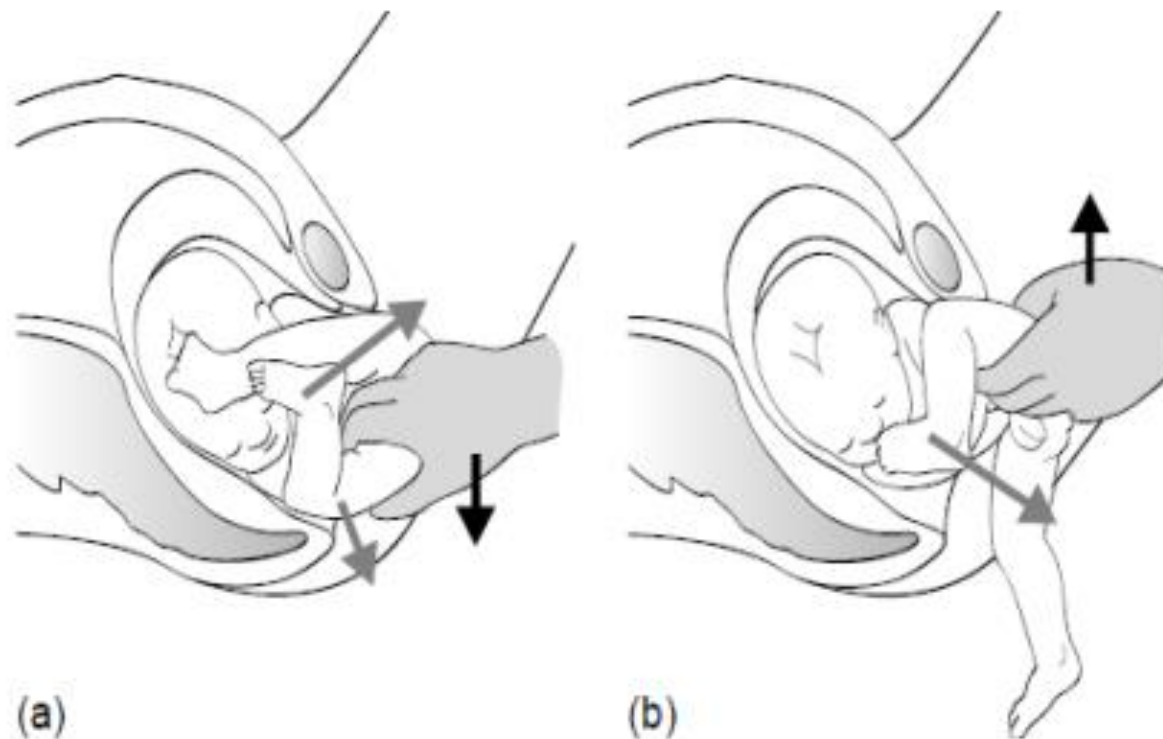
Criteria for allowing vaginal breech delivery:

- Frank or complete breech presentation (footling not for VD) .
- Gestational age > 36 weeks (If preterm : C/S).
- Estimated fetal weight between 2.5 - 3.5kg (big baby for C/S).
- Fetal head must be flexed.
- Adequate maternal pelvis.
- No other obstetric complications.
- Preferably under epidural analgesia.

There should be an operator experienced in vaginal breech delivery available in the hospital.

Intrapartum management of vaginal breech delivery:

- Spontaneous onset of labour is preferred.
- Induction of labour with breech should be only in highly selected cases as C/S may be a better option than Induction.
- Mothers are advised to attend the delivery unit when membranes rupture or with onset of painful contractions.
- Cord presentation or prolapse should be excluded on admission.
- The labour is conducted as for vertex presentation.
- The progress of labour and fetal heart rate are monitored by partogram.
- If progress of labour was poor, adequacy of uterine contractions should be evaluated.
- Limited period of oxytocin augmentation could be of value and safe in selected cases.
- If the progress is poor in the first few hour of augmentation, it is better to opt for C/S.



Delivery of the extended legs by slight abduction of the thigh and flexion at the knees.

-When the scapulae become visible, if the arms are flexed the forearms are delivered by sweeping it in front of the fetal chest.

-If the arms are extended adduction and flexion of the shoulder followed by flexion at the elbow helps to bring down the forearm and hand.

In case this was not possible,

'**Lovset manoeuvre**' is resorted to where the posterior shoulder, which is below the level of the sacral promontory, is brought anterior below the symphysis pubis by rotating the fetus clockwise by holding the baby with the thumbs on the sacrum and index fingers on the anterior



Delivery of the after coming



-After delivery of the shoulder the dorsum of the fetus should face anterior and on vaginal examination the chin should be facing the sacrum.

-The descent of the head in the pelvis is assisted by the weight of the fetus which is gently supported till the nape of the neck is seen under the symphysis pubis.

-This signals that the head is low in the pelvis and could be delivered by

1. Jaw flexion and shoulder traction (Mauriceau–Smellie Veit manoeuvre) can be employed where two fingers of one hand are pressed over the maxilla to flex the head and fingers of other hand put on shoulders to apply shoulder traction and deliver the head.

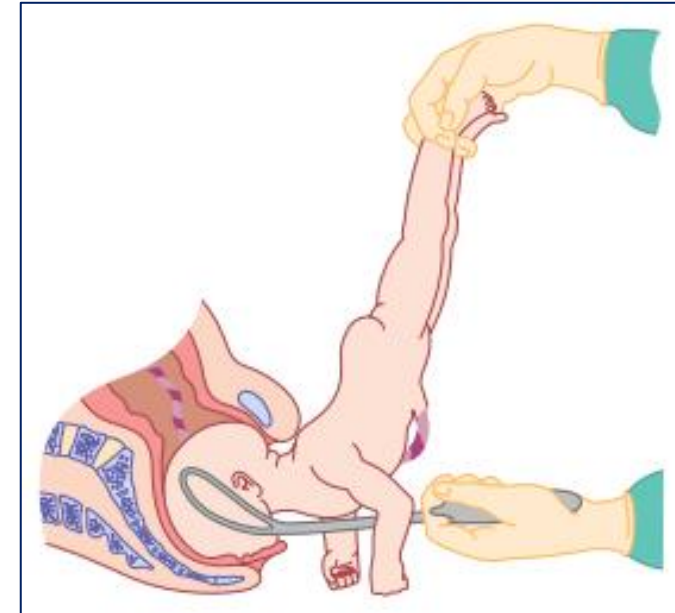


Delivery of the head by jaw flexion and shoulder traction.

2. If this manoeuvre proves difficult, forceps can be used to deliver the head.

- An assistant holds the baby's body upwards while the forceps are applied to the head. .

-The head should be slowly delivered, not allowing any sudden decompression which could result in pressure alterations inside the skull and so cause intra-cerebral venous bleeding.





Breech birth demonstration using the clinical simulation suite.flv

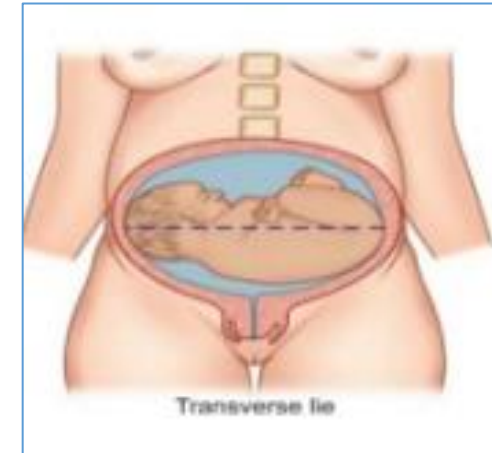
Complications of a breech labour and delivery:

- Increased risk of cord prolapse: particularly with footling breech.
- Mechanical difficulties with the delivery of the shoulders and/or after-coming head, leading to damage of the visceral organs or the brachial plexus.
- Delay in the delivery of the head may occur with a larger fetus, leading to prolonged compression of the umbilical cord and asphyxia.
- Uncontrolled rapid delivery of the head may occur with a smaller fetus and predisposes to tentorial tears and intracranial bleeding.
- A small or preterm fetus may deliver through an incompletely dilated cervix, resulting in head entrapment

Transverse, oblique and unstable lie

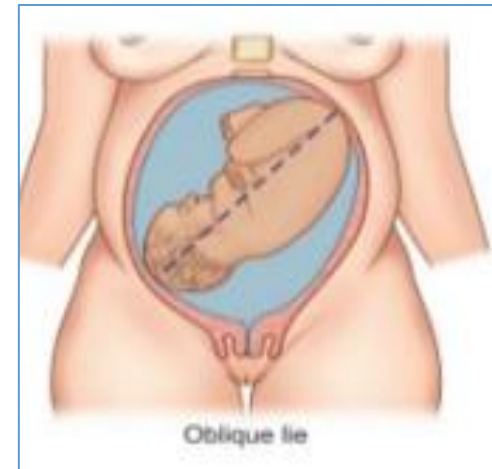
*A transverse lie :

- When the fetal long axis lies perpendicular to that of the maternal long axis and classically results in a shoulder presentation.



*An oblique lie:

- When the long axis of the fetal body crosses the long axis of the maternal body at an angle close to 45°.



*-Unstable lie:

- Defined as a fetus who frequently changes its lie from transverse to longitudinal to oblique after 36 weeks gestation.

Position:

Only two positions described:

1. Dorso-posterior position (the back directed posterior).
2. Dorsoanterior (the back directed anterior).

diagnosis

Abdominal examination:

- The uterus appears asymmetrical, and broader than usual.
- The fundus lies lower than expected for the duration of pregnancy.
- The hard round head is felt in one iliac fossa or flank.
- The breach felt on the other side.
- No presenting part felt over the pelvic brim.

Vaginal examination:

When the cervix become dilated

- An arm or a loop of cord may prolapse ;
- Diagnosis of shoulder presentation depends on recognition of the acromion process, scapula and adjacent ribs.

Course of labour:

- A fetus lying transversely cannot be born vaginally unless it is macerated or very premature.
- There is no true mechanism of labour.
- Untreated case will end in obstructed labour and fetal death.
- So women with transverse lie should not allowed to labour because of the risk of cord prolapse and uterine rupture.

Management after 37weeks:

- All women with transverse or oblique or unstable lie should be instructed to attend for assessment if they begin to contract or suspect to loss liquor.
- Ultrasound assessment should be done.
- If there is no indication for delivery by caesarean section the management either expectant or active.

-Expectant management:

- Waiting for spontaneous version to longitudinal lie which will occur in 80-85 percent of women prior to labour.
- Daily assessment and discharge home once the lie has stabilized longitudinally for 48 hours.

Active management:

1. Trying ECV , either alone and discharge home if successful.
2. Or ECV with induction of labour (stabilizing induction).
3. If transverse or oblique lie persist after 40weeks elective C/S should be done.

Woman who present in labour:

- Very early labour with intact membranes we may try ECV.
- If labour established with ruptured membranes we do emergency C/S.
- In neglected cases when the uterus is tonically retracted there is great risk of uterine rupture C/S should be done even when the fetus is dead.



Thank you